FIBONACCI POSITIVITY OF THE ALTERNATIVE q-CHARLIER SPECIALIZATION

KNOWLEDGE SENSE

Abstract. Petrov and Scott conjectured that the alternative q-Charlier specialization of Okada's biserial clone Schur functions is Fibonacci positive for some values of its two parameters ρ and q. We prove that this holds for all $0 < \rho \le 1/4$ and $0 < q \le 1/5$. More precisely, for every parameter pair in this range, every Fibonacci word w and every integer $s \ge 0$, the clone Schur function of w evaluated at the specialization shifted by s is strictly positive. The proof uses the reduction of Petrov and Scott to a single sequence (t_k). The new input consists of two estimates on t_k and on t_{k+1}/t_k, uniform in k and in the parameters, which give the positivity of the tridiagonal determinants occurring in the recursive definition of the clone Schur functions.

1. Introduction

A *Fibonacci word* is a finite word $w = w_1 \cdots w_r$ with letters $w_i \in \{1, 2\}$; its *weight* is $|w| = w_1 + \cdots + w_r$, and the empty word $\varnothing$ has weight zero. Fibonacci words form the Young–Fibonacci lattice $\mathbb{YF}$, a 1-differential poset introduced by Fomin [2] and Stanley [5]. Okada [3] attached to $\mathbb{YF}$ a family of noncommutative clone Schur functions. Their evaluations $s_w(x \mid y)$ at two sequences $x = (x_k)_{k\ge1}$ and $y = (y_k)_{k\ge1}$, the biserial clone Schur functions, satisfy a clone Pieri rule, so that, when all x_k are nonzero, $w \mapsto s_w(x \mid y)/(x_1 \cdots x_{|w|})$ is a harmonic function on $\mathbb{YF}$ [4, Proposition 2.6].

Petrov and Scott [4] call (x, y) *Fibonacci positive* if $s_w(x \mid y) > 0$ for every Fibonacci word w; such specializations give coherent families of probability measures on Fibonacci words. They characterize Fibonacci positive specializations [4, Theorem 4.12], relate them to Stieltjes moment sequences and to orthogonal polynomials from the q-Askey scheme, and give a number of examples. For real parameters ρ and q, consider the *alternative q-Charlier specialization*

(1.1)
$$x_k = \frac{q^{k-1}\bigl(1 + \rho q^{k-2} + \rho q^{k-1} - \rho q^{2k-2}\bigr)}{(1 + \rho q^{2k-3})(1 + \rho q^{2k-1})}, \qquad y_k = \frac{\rho\, q^{3k-2}(1 - q^k)(1 + \rho q^{k-1})}{(1 + \rho q^{2k})(1 + \rho q^{2k-1})^2(1 + \rho q^{2k-2})},$$

for $k \ge 1$. When $\rho > 0$ and $q > 0$, every denominator in (1.1) is positive, including the factor $1 + \rho q^{-1}$ that occurs at $k = 1$.

Conjecture 1.1 ([4, Conjecture 6.3])**.** *There exist values of ρ and q for which the specialization* (1.1) *is Fibonacci positive.*

For a sequence $a = (a_k)_{k\ge1}$ and an integer $s \ge 0$ we write $a + s$ for the shifted sequence $(a_{k+s})_{k\ge1}$. Our main result is the following.

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Theorem 1.2. *Let $0 < \rho \le 1/4$ and $0 < q \le 1/5$, and let (x, y) be given by (1.1). Then*

$$s_w(x + s \mid y + s) > 0$$

for every Fibonacci word w and every integer $s \ge 0$. In particular, (x, y) is Fibonacci positive, and Conjecture 1.1 holds.

We do not determine the largest set of parameters for which (1.1) is Fibonacci positive; Remark 6.4 records the conditions on (ρ, q) used in the argument.

What is new. The framework is that of Petrov and Scott. Following [4, Section 4.1], one writes the specialization (1.1) as

$$x_k = c_k(1 + t_{k-1}), \qquad y_k = c_k c_{k+1} t_k \qquad (k \ge 1), \qquad t_0 = 0,$$

with positive sequences c and t; these sequences are recorded in [4, Conjecture 6.3] and recalled in (3.1). The rescaling identity [4, (2.13)] reduces positivity to the normalized pair $u_k = 1+t_{k-1}$, $v_k = t_k$, and the expansions of the normalized tridiagonal determinants used in the proof of [4, Proposition 4.7] express the relevant determinants A_ℓ and B_ℓ in terms of t.

The contribution of this paper is a pair of estimates for this particular sequence t: on the stated parameter range,

$$t_k \le \tfrac{5}{16} \qquad \text{and} \qquad \frac{t_{k+1}}{t_k} \le \tfrac{3}{8} \qquad (k \ge 1),$$

uniformly in k, ρ and q (Lemma 4.1). These give, for the normalized determinants B_ℓ, the uniform relative bound

$$B_\ell(u + m \mid v + m) \ge \tfrac{25}{88}\, t_{m+1} \qquad (\ell, m \ge 0)$$

(Proposition 4.2), from which Theorem 1.2 follows by induction on the number of letters 2. To keep the argument self-contained and to treat all shifts at once, we derive the finite determinant identities directly from the definitions; the proof of Theorem 1.2 does not invoke the characterization theorem [4, Theorem 4.12]. In the terminology of [4, Definition 4.6], the estimates show that t is of convergent type (Remark 6.1).

Organization. Section 2 recalls the clone Schur functions, the rescaling identity and the expansions of normalized determinants. Section 3 verifies the factorization of (1.1). Section 4 proves the uniform estimates, and Section 5 proves Theorem 1.2. Section 6 relates the result to the general theory of [4] and to the alternative q-Charlier polynomials studied by Atakishiyev and Klimyk [1].

2. Clone Schur functions

Let $a = (a_k)_{k\ge1}$ and $b = (b_k)_{k\ge1}$ be real sequences. For $\ell \ge 1$ let $A_\ell(a \mid b)$ be the determinant of the $\ell \times \ell$ tridiagonal matrix with diagonal $a_1, \dots, a_\ell$, superdiagonal $b_1, \dots, b_{\ell-1}$ and all subdiagonal entries equal to 1, and put $A_0(a \mid b) = 1$. For $\ell \ge 0$ let $B_\ell(a \mid b)$ be the determinant of the $(\ell + 1) \times (\ell + 1)$ matrix

$$\begin{pmatrix} b_1 & a_1 b_2 & 0 & \cdots & 0 \\ 1 & a_3 & b_3 & \ddots & \vdots \\ 0 & 1 & a_4 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b_{\ell+1} \\ 0 & \cdots & 0 & 1 & a_{\ell+2} \end{pmatrix},$$

so that $B_0(a \mid b) = b_1$. Thus B_ℓ has diagonal $b_1, a_3, \dots, a_{\ell+2}$, superdiagonal $a_1b_2, b_3, \dots, b_{\ell+1}$ and subdiagonal entries 1. In the notation of [4, (2.10)], B_ℓ is the determinant of size $\ell+1$.

Every nonempty Fibonacci word can be written uniquely either as 1^ℓ with $\ell \geq 1$ or as $1^\ell 2z$ with $\ell \geq 0$ and z a Fibonacci word.

Definition 2.1 ([3], [4, Definition 2.5])**.** The biserial clone Schur functions are defined recursively by $s_\varnothing(a \mid b) = 1$ and

$$s_{1^\ell}(a \mid b) = A_\ell(a \mid b), \qquad s_{1^\ell 2z}(a \mid b) = B_\ell\big(a + |z| \;\big|\; b + |z|\big)\, s_z(a \mid b).$$

The recursion terminates because z contains fewer letters 2 than $1^\ell 2z$. A pair (a, b) is *Fibonacci positive* if $s_w(a \mid b) > 0$ for every Fibonacci word w [4, Definition 4.1].

The next identity is [4, (2.13)]. We include the short verification, since we use it for shifted sequences.

Lemma 2.2. *Let $\gamma = (\gamma_k)_{k\geq 1}$ be a sequence of nonzero reals, and put $a'_k = \gamma_k a_k$ and $b'_k = \gamma_k\gamma_{k+1}b_k$. Then*

$$s_w(a' \mid b') = \gamma_1 \cdots \gamma_{|w|}\, s_w(a \mid b)$$

for every Fibonacci word w.

Proof. Let M be an $n \times n$ tridiagonal matrix with diagonal entries d_i, superdiagonal entries e_i and subdiagonal entries 1, and let $g_1, \dots, g_n$ be nonzero reals. Put $G = \operatorname{diag}(g_1, \dots, g_n)$ and let P be the diagonal matrix with entries $p_i = g_1 g_2 \cdots g_i$. The matrix $GP^{-1}MP$ is tridiagonal with diagonal $g_i d_i$, superdiagonal $g_i g_{i+1} e_i$ and subdiagonal entries 1, and its determinant is $g_1 \cdots g_n \det M$.

With $g_i = \gamma_i$ this gives $A_\ell(a' \mid b') = \gamma_1 \cdots \gamma_\ell A_\ell(a \mid b)$. Let $r \geq 0$. For $\ell = 0$ we have $B_0(a' + r \mid b' + r) = b'_{r+1} = \gamma_{r+1}\gamma_{r+2}\, b_{r+1}$. For $\ell \geq 1$, the matrix defining $B_\ell(a' + r \mid b' + r)$ arises in the same way from that of $B_\ell(a + r \mid b + r)$ with $g_1 = \gamma_{r+1}\gamma_{r+2}$ and $g_i = \gamma_{r+i+1}$ for $2 \leq i \leq \ell + 1$; indeed $g_1 g_2 = \gamma_{r+1}\gamma_{r+2}\gamma_{r+3}$ is the factor acquired by the entry $a_{r+1}b_{r+2}$. Hence $B_\ell(a' + r \mid b' + r) = \gamma_{r+1} \cdots \gamma_{r+\ell+2}\, B_\ell(a + r \mid b + r)$ for all $\ell \geq 0$. Since $|1^\ell 2z| = |z| + \ell + 2$, the lemma follows by induction on the number of letters 2 in w. □

Applying Lemma 2.2 to the shifted sequences $a + s$, $b + s$ and $\gamma + s$ gives

$$s_w(a' + s \mid b' + s) = \Big(\prod_{j=s+1}^{s+|w|} \gamma_j\Big)\, s_w(a + s \mid b + s) \qquad (s \geq 0). \tag{2.1}$$

We now fix a real sequence $(t_k)_{k\geq 0}$ with $t_0 = 0$ and $t_k \geq 0$ for $k \geq 1$, and consider the normalized sequences

$$u_k = 1 + t_{k-1}, \qquad v_k = t_k \qquad (k \geq 1).$$

For integers $m, \ell \geq 0$ put

$$D_\ell(m) = A_\ell(u + m \mid v + m), \qquad E_\ell(m) = B_\ell(u + m \mid v + m), \qquad \delta_m = 1 + t_m - t_{m+1}.$$

The matrix defining $D_\ell(m)$ has diagonal entries $1+t_m, \dots, 1+t_{m+\ell-1}$ and superdiagonal entries $t_{m+1}, \dots, t_{m+\ell-1}$. The matrix defining $E_\ell(m)$ has diagonal entries t_{m+1} and $1 + t_{m+2}, \dots, 1 + t_{m+\ell+1}$, and superdiagonal entries $(1+t_m)t_{m+2}$ and $t_{m+3}, \dots, t_{m+\ell+1}$. The following expansions appear in the proof of [4, Proposition 4.7].

Lemma 2.3. *For all $m, \ell \geq 0$,*

$$D_\ell(m) = 1 + \sum_{j=1}^{\ell} t_m t_{m+1} \cdots t_{m+j-1}. \tag{2.2}$$

Moreover $E_0(m) = t_{m+1}$*, and for* $\ell \geq 1$*,*

$$E_\ell(m) = t_{m+1} - \delta_m\, t_{m+2}\, D_{\ell-1}(m+3). \tag{2.3}$$

Proof. We have $D_0(m) = 1$ and $D_1(m) = 1 + t_m$. Expansion along the last row gives

$$D_\ell(m) = (1 + t_{m+\ell-1})D_{\ell-1}(m) - t_{m+\ell-1}D_{\ell-2}(m) \qquad (\ell \geq 2),$$

so $D_\ell(m) - D_{\ell-1}(m) = t_{m+\ell-1}\big(D_{\ell-1}(m) - D_{\ell-2}(m)\big)$. By induction $D_\ell(m) - D_{\ell-1}(m) = t_m t_{m+1} \cdots t_{m+\ell-1}$ for $\ell \geq 1$, which gives (2.2).

The value $E_0(m) = v_{m+1} = t_{m+1}$ is immediate. Let $\ell \geq 1$. Deleting the first row and the first column of the matrix defining $E_\ell(m)$ leaves the matrix of $D_\ell(m+2)$, and deleting the first two rows and columns leaves that of $D_{\ell-1}(m+3)$. Expansion along the first row therefore gives

$$E_\ell(m) = t_{m+1}D_\ell(m+2) - (1 + t_m)t_{m+2}D_{\ell-1}(m+3).$$

By (2.2), $D_\ell(m+2) = 1 + t_{m+2}D_{\ell-1}(m+3)$, also when $\ell = 1$. Substituting this proves (2.3). □

3. The alternative q-Charlier specialization

Let $\rho > 0$ and $0 < q < 1$. Following [4, Conjecture 6.3], define for $k \geq 1$

$$c_k = \frac{q^{k-1}(1 + \rho q^{k-1})}{(1 + \rho q^{2k-1})(1 + \rho q^{2k-2})}, \qquad t_k = \frac{\rho\, q^{k-1}(1 - q^k)(1 + \rho q^{2k+1})}{(1 + \rho q^k)(1 + \rho q^{2k-1})}, \tag{3.1}$$

and put $t_0 = 0$. Clearly $c_k > 0$ and $t_k > 0$ for $k \geq 1$. In [4] these sequences are obtained from (1.1) through [4, Proposition 4.4]; we verify the factorization directly.

Lemma 3.1. *For* $\rho > 0$*,* $0 < q < 1$ *and every* $k \geq 1$*,*

$$x_k = c_k(1 + t_{k-1}), \qquad y_k = c_k c_{k+1} t_k.$$

Proof. For $k = 1$, the factor $1 + \rho q^{-1}$ cancels in (1.1) and $x_1 = 1/(1+\rho q) = c_1 = c_1(1+t_0)$. Let $k \geq 2$. Bringing $c_k(1 + t_{k-1})$ to the common denominator $(1 + \rho q^{2k-1})(1 + \rho q^{2k-2})(1 + \rho q^{2k-3})$ gives the numerator $q^{k-1}N_k$, where

$$\begin{aligned} N_k &= (1 + \rho q^{k-1})(1 + \rho q^{2k-3}) + \rho q^{k-2}(1 - q^{k-1})(1 + \rho q^{2k-1}) \\ &= 1 + \rho q^{k-2} + \rho q^{k-1} + \rho^2 q^{3k-4} + \rho^2 q^{3k-3} - \rho^2 q^{4k-4} \\ &= (1 + \rho q^{2k-2})\big(1 + \rho q^{k-2} + \rho q^{k-1} - \rho q^{2k-2}\big). \end{aligned}$$

Cancelling the factor $1 + \rho q^{2k-2}$ yields the formula for x_k in (1.1). In the product $c_k c_{k+1} t_k$ the factors $1 + \rho q^k$ and $1 + \rho q^{2k+1}$ cancel, the power of q is $q^{(k-1)+k+(k-1)} = q^{3k-2}$, and what remains is the expression for y_k in (1.1). □

By Lemma 3.1 and (2.1) with $\gamma = c$,

$$s_w(x + s \mid y + s) = \Big(\prod_{j=s+1}^{s+|w|} c_j\Big)\, s_w(u + s \mid v + s) \qquad (s \geq 0), \tag{3.2}$$

where $u_k = 1 + t_{k-1}$ and $v_k = t_k$ are built from the sequence (3.1).

4. Uniform estimates

Throughout this section $0 < \rho \le 1/4$ and $0 < q \le 1/5$, and t is the sequence (3.1). Put

$$T = \frac{5}{16}, \qquad R = \frac{3}{8}.$$

Lemma 4.1. *For every $k \ge 1$,*

$$0 < t_k \le \rho(1+\rho)\, q^{k-1} \le T \qquad \text{and} \qquad 0 < \frac{t_{k+1}}{t_k} \le q(1+q)(1+\rho)^2 \le R.$$

Proof. In (3.1) we have $1 - q^k < 1$, $1 + \rho q^{2k+1} \le 1 + \rho$ and the denominator is at least 1; hence $t_k \le \rho(1+\rho)q^{k-1} \le \frac{1}{4} \cdot \frac{5}{4} = T$. Dividing the expressions (3.1) for t_{k+1} and t_k gives

$$\frac{t_{k+1}}{t_k} = q \cdot \frac{1-q^{k+1}}{1-q^k} \cdot \frac{1+\rho q^{2k+3}}{1+\rho q^{2k+1}} \cdot \frac{1+\rho q^k}{1+\rho q^{k+1}} \cdot \frac{1+\rho q^{2k-1}}{1+\rho q^{2k+1}}.$$

The inequality $(1-q^{k+1})/(1-q^k) \le 1+q$ is equivalent to $q^k \le q$, which holds for $k \ge 1$. The second fraction is at most 1, and each of the last two is at most $1+\rho$, since its numerator is at most $1+\rho$ and its denominator is at least 1. Therefore

$$\frac{t_{k+1}}{t_k} \le q(1+q)(1+\rho)^2 \le \frac{1}{5} \cdot \frac{6}{5} \cdot \frac{25}{16} = R. \qquad \square$$

Proposition 4.2. *For all integers $m, \ell \ge 0$,*

$$1 \le D_\ell(m) \le \frac{16}{11} \qquad \text{and} \qquad E_\ell(m) \ge \frac{25}{88}\, t_{m+1} > 0.$$

Proof. By Lemma 4.1, each product $t_m t_{m+1} \cdots t_{m+j-1}$ in (2.2) is nonnegative and at most T^j. Hence $1 \le D_\ell(m) \le \sum_{j \ge 0} T^j = (1-T)^{-1} = 16/11$.

For $\ell = 0$ we have $E_0(m) = t_{m+1}$. Let $\ell \ge 1$. Since $0 \le t_m, t_{m+1} \le T$, we have $0 < 1 - T \le \delta_m \le 1 + T = 21/16$. Lemma 4.1 with $k = m+1$ gives $t_{m+2} \le R\, t_{m+1}$. By (2.3) and the bound on $D_{\ell-1}(m+3)$,

$$E_\ell(m) \ge t_{m+1} - \frac{21}{16} \cdot \frac{3}{8} \cdot \frac{16}{11}\, t_{m+1} = \Big(1 - \frac{63}{88}\Big) t_{m+1} = \frac{25}{88}\, t_{m+1}.$$

Since $t_{m+1} > 0$, the right-hand side is positive. $\square$

The lower bound for $E_\ell(m)$ in Proposition 4.2 is proportional to t_{m+1}. No positive lower bound uniform in m is possible, since $E_0(m) = t_{m+1} \to 0$ as $m \to \infty$.

5. Proof of the main theorem

Proof of Theorem 1.2. Fix $s \ge 0$, and let $u_k = 1 + t_{k-1}$, $v_k = t_k$ with t as in (3.1). We first show that $s_w(u + s \mid v + s) > 0$ for every Fibonacci word w, by induction on the number of letters 2 in w.

If $w = \varnothing$, then $s_w = 1$. If $w = 1^\ell$ with $\ell \ge 1$, then $s_w(u+s \mid v+s) = A_\ell(u+s \mid v+s) = D_\ell(s) \ge 1$ by Proposition 4.2. If $w = 1^\ell 2 z$, then by Definition 2.1

$$s_w(u+s \mid v+s) = B_\ell\big(u+s+|z| \;\big|\; v+s+|z|\big)\, s_z(u+s \mid v+s) = E_\ell\big(s+|z|\big)\, s_z(u+s \mid v+s).$$

The first factor is positive by Proposition 4.2 with $m = s + |z|$, and the second is positive by the induction hypothesis, since z has fewer letters 2 than w.

Finally, each c_j is positive, so (3.2) gives $s_w(x+s \mid y+s) > 0$. The case $s = 0$ is the Fibonacci positivity of (x, y), which proves Conjecture 1.1. $\square$

The proof gives the explicit lower bound

$$s_w(u+s \mid v+s) \geq \prod_i \frac{25}{88}\, t_{s+|z_i|+1},$$

where $w = 1^{\ell_1}2z_1$, $z_1 = 1^{\ell_2}2z_2$, and so on, the product running over the occurrences of the letter 2 in w.

6. Remarks

Remark 6.1 (Convergent type). Let

$$A_\infty(m) = 1 + \sum_{j\geq 1} t_m t_{m+1}\cdots t_{m+j-1}, \qquad B_\infty(m) = t_{m+1} - \delta_m\, t_{m+2}\, A_\infty(m+3)$$

be the quantities of [4, (4.4)]. Under the hypotheses of Theorem 1.2, the proof of Proposition 4.2 shows that $A_\infty(m) \leq 16/11$ and $B_\infty(m) \geq \frac{25}{88} t_{m+1} > 0$ for all $m \geq 0$. Thus t is a sequence of convergent type in the sense of [4, Definition 4.6]. Fibonacci positivity of the normalized pair (u, v) can therefore also be deduced from [4, Proposition 4.7], that of (x, y) then follows from the rescaling identity [4, (2.13)], and the positivity of the shifted specializations follows from [4, Proposition 5.8].

Remark 6.2 (Type-I component). By Lemma 4.1, $\sum_{k\geq 1} t_k \leq \rho(1+\rho)/(1-q) < \infty$, so the product $\prod_{k\geq 1}(1+t_k)$ converges and the Type-I component of the clone harmonic function of (x, y) satisfies $\mu_{\mathrm{I}}(1^\infty) = \prod_{k\geq 1}(1+t_k)^{-1} > 0$ [4, Lemma 15.3]. Since t is of convergent type, [4, Propositions 15.4 and 15.6] apply: $\mu_{\mathrm{I}}(1^\infty 2w) > 0$ for every Fibonacci word w, and μ_{I} is a probability measure on the Type-I words. By [4, Theorem 1.5], a random Fibonacci word of weight n distributed according to the clone coherent measure of (x, y) converges in distribution, as $n \to \infty$, to a random Type-I word with law μ_{I}, that is, to 1^∞ or to $1^\infty 2v$ with v a finite Fibonacci word.

Remark 6.3 (Orthogonal polynomials). Let $a = \rho > 0$ and $0 < q < 1$, and let

$$\alpha_n = q^n \frac{1+aq^n}{(1+aq^{2n})(1+aq^{2n+1})}, \qquad \beta_n = aq^{2n-1}\frac{1-q^n}{(1+aq^{2n-1})(1+aq^{2n})} \qquad (n \geq 0)$$

be the coefficients, denoted A_n and C_n in [1, Section 2], of the three-term recurrence of the alternative q-Charlier polynomials

$$K_n(\lambda; a; q) = {}_2\phi_1(q^{-n}, -aq^n; 0; q, q\lambda).$$

Comparing with (3.1), we have $\alpha_n = c_{n+1}$ and $\beta_n = c_{n+1}t_n$ for $n \geq 0$; in particular $\beta_0 = 0$. Lemma 3.1 therefore gives

$$x_{n+1} = \alpha_n + \beta_n \quad (n \geq 0), \qquad y_n = \alpha_{n-1}\beta_n \quad (n \geq 1).$$

Consequently the monic polynomials P_n of [4, Theorem 7.1], defined by $P_0 = 1$, $P_1(\lambda) = \lambda - x_1$ and

$$P_{n+1}(\lambda) = (\lambda - x_{n+1})P_n(\lambda) - y_n P_{n-1}(\lambda) \qquad (n \geq 1),$$

are $P_n = (-1)^n \alpha_0\alpha_1\cdots\alpha_{n-1} K_n(\,\cdot\,; \rho; q)$. Atakishiyev and Klimyk [1, (10) and Proposition in Section 3] show that the K_n are orthogonal with respect to the discrete measure with mass proportional to $\rho^n q^{n(n+1)/2}/(q;q)_n$ at q^n, $n \geq 0$. Since x and y are bounded, the corresponding moment problem is determinate, and this measure, normalized to total mass one, is the measure ν attached to (x, y) by [4, Theorem 7.1].

Two comments are in order. First, for the Al-Salam–Chihara and q-Charlier specializations of [4], Fibonacci positivity leads to orthogonal polynomials whose deformation parameter exceeds one [4, Sections 1.3 and 10]. Here the orthogonal polynomials are the alternative q-Charlier polynomials with $0 < q < 1$, the range considered in [1]. Second, by [4, Theorem 7.1], the existence of a nonnegative measure on $[0, \infty)$ with infinite support and with given recurrence coefficients is equivalent to the total positivity of the tridiagonal matrix with diagonals $(1, 1, \dots)$, (x_k) and (y_k). This condition requires $y_k > 0$ and $A_\ell(x + m \mid y + m) > 0$ for all $\ell, m \geq 0$, but it does not involve the determinants B_ℓ. Fibonacci positivity is strictly stronger: the Laguerre polynomials are orthogonal with respect to a measure on $[0, \infty)$, but their recurrence coefficients (x, y) give $B_2(x + 1 \mid y + 1) < 0$ [4, Section 10.6.2]. For this reason the estimates of Section 4 are needed for the determinants B_ℓ. The results of [1] are not used in the proof of Theorem 1.2.

Remark 6.4 (Parameter range). Let $\rho > 0$ and $0 < q < 1$, and put

$$T' = \rho(1 + \rho), \qquad R' = q(1 + q)(1 + \rho)^2.$$

For these values of ρ and q, the calculations in the proof of Lemma 4.1 give, for every $k \geq 1$,

$$t_k \leq T' q^{k-1}, \qquad \frac{t_{k+1}}{t_k} \leq R'.$$

In particular the series $A_\infty(m)$ of Remark 6.1 converge for all m. By [4, Proposition 4.7] and the rescaling identity [4, (2.13)], the specialization (1.1) is therefore Fibonacci positive if and only if $B_\infty(m) \geq 0$ for all $m \geq 0$.

Assume in addition that

$$T' < 1 \qquad \text{and} \qquad (1 + T')R' < 1 - T'.$$

Then the argument of Sections 4 and 5 applies with T, R and $16/11$ replaced by T', R' and $(1 - T')^{-1}$. Hence the conclusion of Theorem 1.2 holds for every such pair (ρ, q), with lower bound $\big(1 - (1 + T')R'/(1 - T')\big)t_{m+1}$ in Proposition 4.2. The rectangle of Theorem 1.2 is a convenient subset of this region with simple constants. An explicit description, in terms of ρ and q, of the region where (1.1) is Fibonacci positive remains open.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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